Master Math for JEE Main & JEE Advanced

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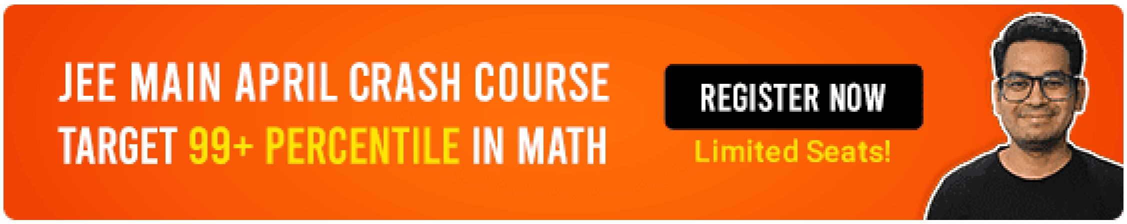
For JEE Main 2020 April



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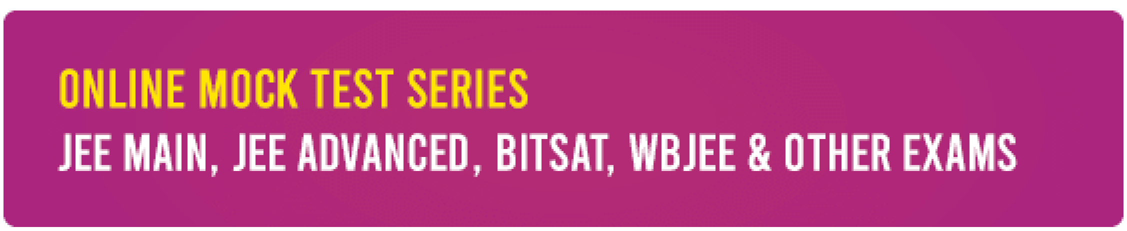

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JEE Mains 2020 Jan Chapter wise Question Bank

Thermal Properties of Matter

Q1

1 litre of a gas at STP is expanded adiabatically to 3 litre. Find work done by the gas. Given γ = 1.40 and $3^{1.4}$= 4.65

एक गैस के 1 लीटर को STP पर 3 लीटर तक प्रसारित किया जाता है। गैस द्वारा किया गया कार्य होगा। दिया है γ = 1.40 तथा $3^{1.4}$= 4.65

(1) 100.8J (2) 90.5J (3) 45J (4) 18J

7th Jan Morning

Sol

(2)

P_1 = 1atm, T_1 = 273K

$P_1V_1^{\gamma} = P_2V_2^{\gamma}$

$P_2 = P_1\left[\frac{V_1}{V_2}\right]^{\gamma}$

$= 1\text{atm}\left(\frac{1}{3}\right)^{1.4}$

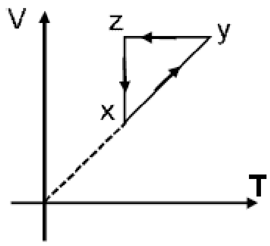
now work done अब किया गया कार्य = $\frac{P_1V_1 - P_2V_2}{\gamma - 1}$ = 88.7 J

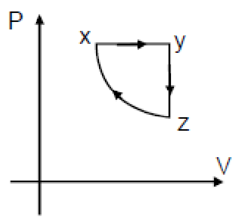
Closest ans is 90.5 J

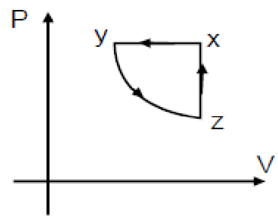
Q2

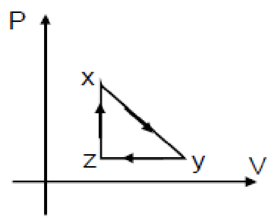
Choose the correct P-V graph of ideal gas for given V-T graph.

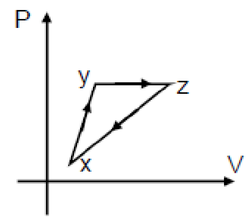
आदर्श गैस के लिए V-T ग्राफ दिया गया है तो इसके संगत P–V ग्राफ बताइए।



(1) 

(2) 

(3) 

(4) 

8th Jan Morning

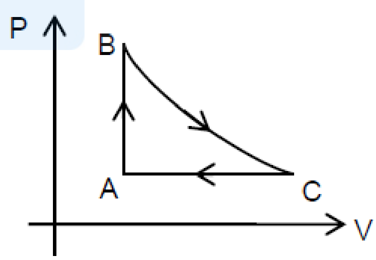
Sol

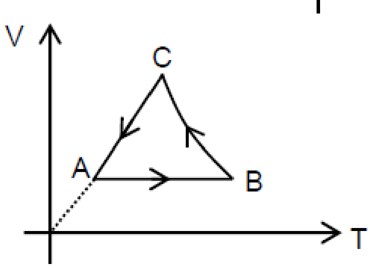
(1)

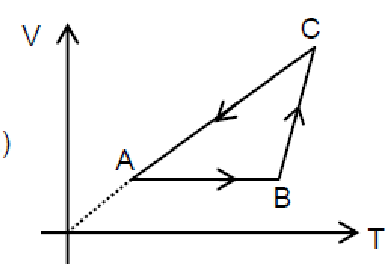
Q3

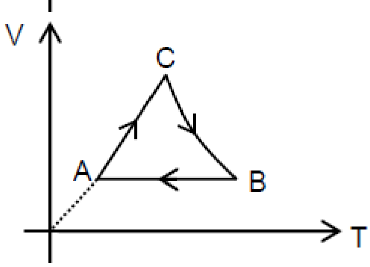
For the given P-V graph for an ideal gas, chose the correct V- T graph. Process BC is adiabatic.

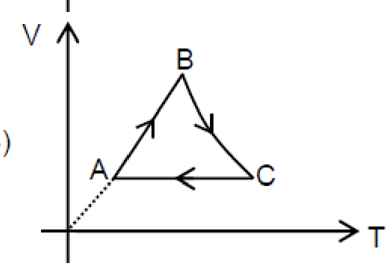
एक आदर्श गैस का P-V ग्राफ दिया गया है। तब सही V-T ग्राफ क्या होगा। प्रक्रम BC रूद्धोष्म है।



(1) 

(2) 

(3) 

(4) 

9^{th} Jan Morning

Sol

(1)

For process A – B प्रक्रम के लिए; Volume is constant आयतन नियत है;

$PV = nRT$; as P increases बढ़ता है; T increases बढ़ता है

For process B – C प्रक्रम के लिए;

PV^{γ} = Constant नियत;

$\Rightarrow TV^{\gamma-1}$ = Constant नियत

For process C – A प्रक्रम के लिए; pressure is constant दाब नियत है

$V = kT$

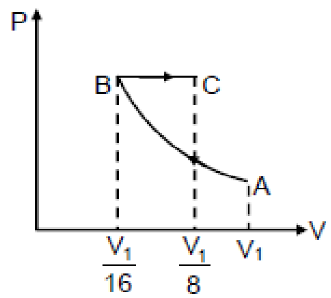
Q4

An ideal gas at initial temperature 300 K is compressed adiabatically (γ = 1.4) to $\left(\frac{1}{16}\right)^{th}$ of its initial volume. The gas is then expanded isobarically to double its volume. Then final temperature of gas round to nearest integer is:

9^{th} Jan Evening

Sol

1819 K



PV^{γ} = constant नियत

$TV^{\gamma-1}$ = constant नियत

$300\,(V_1)^{1.4-1} = T_B\left(\frac{V_1}{16}\right)^{2/5}$

$T_B = 300 \times 2^{8/5}$

Now for BC process

BC प्रक्रम के लिये

$\frac{V_B}{T_B} = \frac{V_c}{T_c}$

$T_c = \frac{V_c T_B}{V_B} = 2 \times 300 \times 2^{8/5}$

T_c = 1819 K

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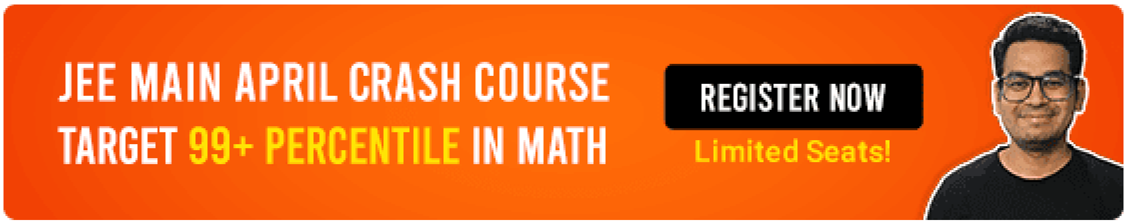
For JEE Main 2020 April



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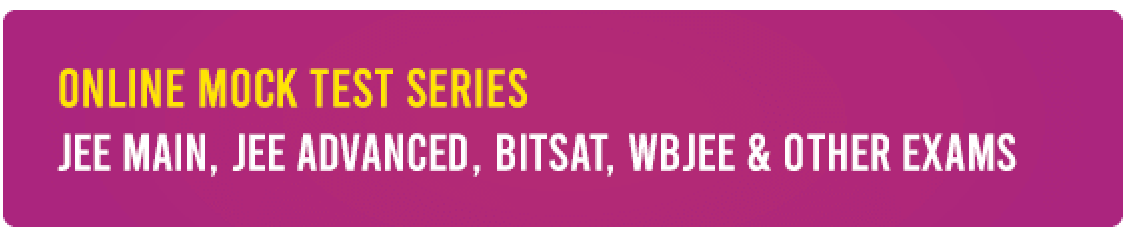

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